

Astro 142

Discussion 4

2025-10-08
[Your name here]

Short Answer Questions

When should you use a dictionary vs a structured array?

Use a dictionary when you have a small number of labeled data fields and don't need vectorized operations (e.g., quick lookups, metadata). Use a structured array when you want NumPy's fast, vectorized operations on multiple named fields (e.g., columns of a table).

When should you use multiple 1D arrays vs a structured array?

Use multiple 1D arrays if fields are manipulated independently or have different lengths. Use a structured array if fields are part of a single dataset (same length) and you want to treat them like columns of one table.

Discussion Activity

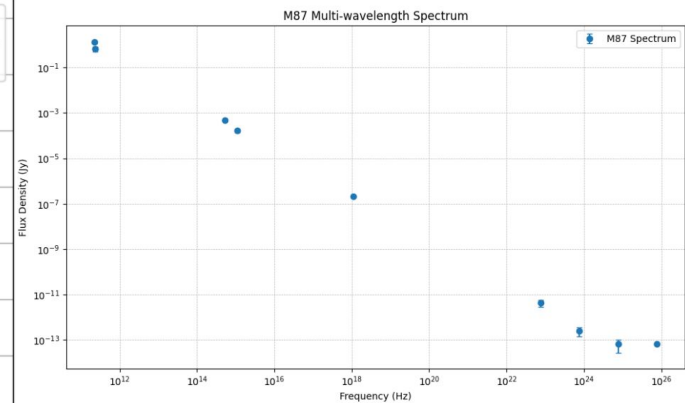
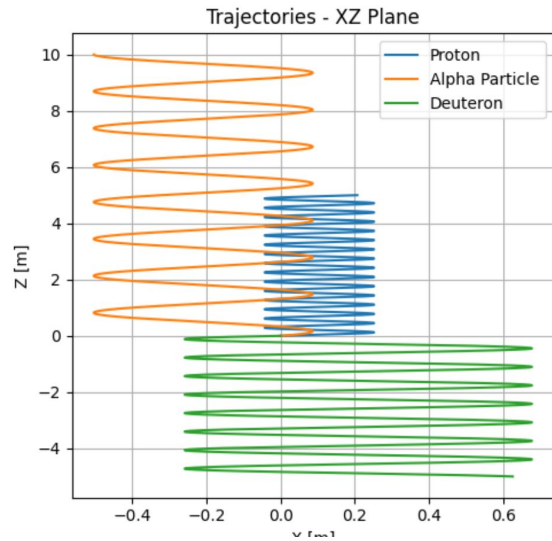
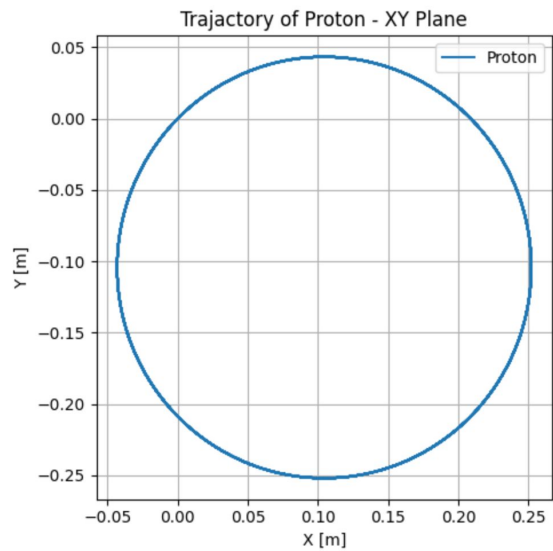
Discussion 4 Activity

Download the python script from Bruinlearn. There are two activities inside.

- Activity 1:
 - Examine the code. This is a simplified version of the cyclotron demo in discussion. How are the particles defined? How are the trajectories calculated? How to plot the trajectories in 2D?
 - Define a new particle deuteron (1 proton+1 neutron).
 - Modify the code to plot the x, y positions of all three particles with proper labels.
 - Make a new plot to plot the x, z positions of all three particles with proper labels.
- Activity 2:
 - Organize the multi-wavelength flux density data of M87 into a proper data structure.
 - Complete the code to plot flux density vs. frequency.

Attach screenshot of code and output on next slide(s).

Start homework 2 when you're done



```
# add a deuteron that starts at the origin and give it a
```

```
particles = [{
    "label": "Proton",
    "q": 1.6e-19,
    "m": 1.67e-27,
    "v0": np.array([1e7, 1e7, 0.5e7]),
    "r0": np.array([0, 0, 0])
},
{
    "label": "Alpha Particle",
    "q": 2*1.6e-19,
    "m": 4*1.67e-27,
    "v0": np.array([1e7, -1e7, 1e7]),
    "r0": np.array([0, 0, 0])
},
{
    "label": "Deuteron",
    "q": 1.6e-19,
    "m": 3.35e-27,
    "v0": np.array([-2e7, 1e7, -0.5e7]),
    "r0": np.array([0, 0, 0])
}]

# ==== Plot trajectory ====
# modify the following code to plot trajectory of all three particles
# create a plot of trajectory in XY
fig_xy = plt.figure(figsize=(5,5))
p1 = particles[0]
plt.plot(p1['r'][:,0], p1['r'][:,1],
         label=f'{p1['label']}' )
plt.title(f'Trajectory of {p1['label']} - XY Plane')
plt.xlabel('X [m]')
plt.ylabel('Y [m]')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

# create a plot of trajectory in XZ

fig_xz = plt.figure(figsize=(5,5)) # ← new figure
for p in particles:
    plt.plot(p['r'][:,0], p['r'][:,2], label=p['label']) # ← plot X vs Z
plt.title('Trajectories - XZ Plane')
plt.xlabel('X [m]')
plt.ylabel('Z [m]')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```



```

# Columns: band minimum (Hz), band center (Hz), band maximum (Hz), flux density (Jy), flu
data = np.array([
    # ALMA
    [2.21e11, 2.21e11, 2.21e11, 1.30e0, 1.30e-1],
    # EHT
    [2.29e11, 2.29e11, 2.29e11, 6.60e-1, 1.60e-1],
    # HST
    [5.18e14, 5.19e14, 5.20e14, 4.91e-4, 0.40e-4],
    [1.09e15, 1.10e15, 1.11e15, 1.64e-4, 0.24e-4],
    # Swift/XRT
    [4.84e17, 1.08e18, 2.42e18, 2.08e-7, 0.00e0],
    # Fermi/LAT
    [2.42e22, 7.65e22, 2.42e23, 4.27e-12, 1.32e-12],
    [2.42e23, 7.65e23, 2.42e24, 2.45e-13, 9.83e-14],
    [2.42e24, 7.65e24, 2.42e25, 6.49e-14, 3.87e-14],
    [2.42e25, 7.65e25, 2.42e26, 6.88e-14, 0.00e0],
])

band_center = data[:,1]
flux = data[:,3]
flux_err = data[:,4]

# Plot the spectrum
fig = plt.figure(figsize=(10, 6))

plt.errorbar(band_center, flux, yerr=flux_err, fmt='o', capsize=3, label='M87 Spectrum')
plt.xscale('log')
plt.yscale('log')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Flux Density (Jy)')
plt.title(['M87 Multi-wavelength Spectrum'])
plt.grid(True, which='both', linestyle='--', linewidth=0.5)
plt.legend()
plt.tight_layout()
plt.show()

```